

OSSP:

Commands:

uname: Displays the operating system and kernel information.

lscpu: Displays processor details like CPU model, cores, and architecture.

```
balaji@BalajiPatil:~$ nano balaji.c
balaji@BalajiPatil:~$ gcc balaji.c -o balaji
balaji@BalajiPatil:~$ ./balaji
Enter a Linux command: uname

Parent Process
Parent PID : 500
Child PID  : 501

Child Process
Child PID : 501
Parent PID: 500

Executing command: uname
```

ps: Lists all active processes currently running in the system.

```
balaji@BalajiPatil:~$ ./balaji
Enter a Linux command: ps

Parent Process
Parent PID : 515
Child PID  : 516

Child Process
Child PID : 516
Parent PID : 515

Executing command: ps

  PID TTY          TIME CMD
  335 pts/0        00:00:00 bash
  515 pts/0        00:00:00 balaji
  516 pts/0        00:00:00 ps
```

Relation Between Hardware Resources and Operating System Services:

1. CPU

The operating system manages CPU usage by sharing processor time among different programs.

Example: Running a browser and a media player at the same time.

2. Memory (RAM) -> Memory Management

The OS assigns RAM to applications and prevents them from accessing each other's memory.

Example: VS Code and Chrome use separate memory while running.

3. Storage (SSD) → File System Management

The OS manages files and folders, allowing users to save and access data easily.

Example: Saving a project file on the desktop.

4. I/O Devices → Device Management

The OS uses drivers to communicate with hardware devices and handle input/output operations.